

1 Highlights

2 **DSCFormer: Topology-Aware Crack and Spalling Segmentation in** 3 **Concrete Dams via Dynamic Snake Convolution and Dual-Teacher** 4 **Distillation**

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- 6 • Dynamic Snake Convolution improves crack connectivity by +6.6 ConnR
7 points.
- 8 • Dual-teacher distillation boosts mIoU by +1.9; confidence-aware KD fur-
9 ther improves crack BF1 by +0.8, all at zero inference overhead.
- 10 • SRL benefits teacher training but harms student under distillation.
- 11 • DTKD reduces cross-seed variance by $\sim 3\times$, acting as a training regulariser.
- 12 • DSCConv enables 18.7% cross-dataset mIoU vs. 0.1% for vanilla SegFormer.

DSCFormer: Topology-Aware Crack and Spalling Segmentation in Concrete Dams via Dynamic Snake Convolution and Dual-Teacher Distillation

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Abstract

Automated semantic segmentation of cracks and spalling on concrete dam surfaces is essential for structural health monitoring. While pixel-level metrics such as IoU capture overall coverage, they do not guarantee the *structural continuity* of detected defects—a fragmented crack prediction may achieve adequate IoU while failing to reveal the true crack path, leading to underestimated severity in safety assessment. Existing methods optimise pixel-level objectives without explicitly preserving this continuity. We propose **DSCFormer**, a topology-aware segmentation framework incorporating two complementary enhancements to a SegFormer-B2 backbone: (1) a *Dynamic Snake Convolution* (DSConv) branch that captures elongated crack geometry through deformable sampling, adding only 4M parameters (+14.5%); and (2) *Class-Conditional Dual-Teacher Knowledge Distillation* (DTKD) that transfers complementary knowledge from a task-specific teacher and a foundation-model teacher with per-class weighting and pixel-level confidence modulation based on teacher agreement, *at zero additional inference cost*. On the **DamSegment** dataset (Gharehbaghi et al., 2026) (1,500 images, 3 classes), DSCFormer+DTKD achieves 72.3% foreground mIoU (72.7% with TTA), the highest among all compared methods, with 86.4% centreline Dice. Ablation experiments confirm complementary gains: DSConv drives crack connectivity ($\text{ConnR}_{\text{fg}} +6.6$), while DTKD drives pixel-level accuracy ($\text{mIoU}_{\text{fg}} +1.9$ over DSConv alone) at a modest cost to crack connectivity ($\text{ConnR}_{\text{cr}} -2.6$), with improvements concentrated on Hard-tier images. We also investigate Skele-

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ton Recall Loss (SRL), finding that while it improves crack boundary accuracy in isolation, it degrades the overall pipeline when combined with DTKD (-1.1 mIoU_{fg}), revealing a negative interaction between skeleton-level supervision and soft-label distillation. Multi-seed experiments reveal that DTKD also acts as a strong regulariser, reducing cross-seed variance by $\sim 3\times$ compared to the baseline (mIoU_{fg} std: 0.52 vs. 1.47). Class-conditional teacher weighting provides marginal gains over equal weighting ($+0.1$ mIoU_{fg}) but improves crack centre-line preservation (clDice_{fg} $+0.6$). Zero-shot cross-dataset evaluation on S2DS demonstrates that DSCFormer+DTKD achieves 18.7% mIoU_{fg}, compared to 0.1% for SegFormer-B2 and 2.5% for Mask2Former, indicating that the geometric inductive bias substantially improves cross-domain robustness.

19 *Keywords:* Crack segmentation, Concrete dam inspection, Dynamic snake
 20 convolution, Knowledge distillation, Confidence-aware distillation, Structural
 21 health monitoring

22 1. Introduction

23 Concrete dams are safety-critical infrastructure whose structural integrity
 24 must be continuously monitored throughout multi-decade service lives. Surface
 25 defects—predominantly cracks and spalling—serve as early indicators of internal
 26 deterioration, yet their manual inspection remains labour-intensive, subjective,
 27 and logistically constrained by the scale and accessibility of dam faces (Chen
 28 et al., 2024; Fang and Liu, 2024). Automated semantic segmentation of these
 29 defects from high-resolution imagery therefore promises substantial gains in in-
 30 spection efficiency and repeatability.

31 Recent deep-learning methods have advanced pixel-level defect segmentation
 32 considerably. Encoder-decoder architectures such as U-Net (Ronneberger et al.,
 33 2015) and hierarchical transformers like SegFormer (Xie et al., 2021) achieve
 34 competitive Intersection-over-Union (IoU) on benchmark crack datasets (Zou
 35 et al., 2019; Liu et al., 2019a). However, IoU and related region-overlap metrics
 36 are insensitive to topological errors: a prediction that fragments a continuous

crack into disconnected segments may still score favourably on IoU while misrepresenting the structural extent of the defect (Pantoja-Rosero et al., 2024). For structural safety assessment, the *connectivity* of a predicted crack is as important as its pixel coverage—a fragmented prediction can lead an engineer to underestimate propagation length and thus misjudge residual capacity.

Topology-preserving losses such as cDice (Shit et al., 2021) and persistent-homology penalties (Hu et al., 2019; Clough et al., 2020) partially address this gap, yet they either incur prohibitive GPU memory for high-resolution dam images or operate on binary masks without multi-class support. The recently proposed Skeleton Recall Loss (SRL) (Shit et al., 2024) overcomes both limitations by computing recall on CPU-extracted ground-truth skeletons with over 90% reduction in computational overhead. Concurrently, Dynamic Snake Convolution (DSConv) (Qi et al., 2023) reshapes kernel sampling grids to follow elongated, tortuous paths—a morphological prior that aligns naturally with the geometry of cracks.

Meanwhile, large vision foundation models such as SAM (Kirillov et al., 2023) and DINOv2 (Oquab et al., 2024) offer powerful pretrained representations, yet our experiments show that parameter-efficient fine-tuning via LoRA (Hu et al., 2022) on small industrial datasets underperforms fully fine-tuned task-specific architectures. A natural question arises: *can the complementary strengths of task-specific and foundation models be combined?* We address this through knowledge distillation, where a student model learns from both a task-specific teacher and a foundation-model teacher with class-conditional weighting.

In this paper we propose **DSCFormer**, a framework that augments SegFormer-B2 with a parallel DSConv branch and class-conditional dual-teacher knowledge distillation (DTKD). We additionally investigate Skeleton Recall Loss (SRL) and uncover an asymmetric role in distillation: SRL enriches teacher training but harms the student when combined with DTKD. We evaluate on the semantic segmentation subset of the DamSegment dataset (Gharehbaghi et al., 2026), comprising 1,500 annotated dam surface images. Our contributions are:

- 68 1. A topology-aware segmentation framework with two complementary en-
69 hancements: a parallel Dynamic Snake Convolution branch that drives
70 crack connectivity ($\text{ConnR}_{\text{fg}} + 6.6$) through geometry-aware sampling, and
71 a dual-teacher knowledge distillation scheme (DTKD) that drives pixel-
72 level accuracy ($\text{mIoU}_{\text{fg}} + 1.9$ over DSConv alone) by transferring comple-
73 mentary knowledge from a task-specific teacher and a foundation-model
74 teacher. DTKD incorporates confidence-aware distillation that modulates
75 the KD loss at pixel level based on teacher agreement, improving bound-
76 ary quality ($\text{BF1}_{\text{cr}} + 0.8$, $\text{cIDice}_{\text{cr}} + 0.6$) at no additional inference cost.
77 The framework adds only 4M parameters with no inference-time overhead
78 from distillation.
- 79 2. An empirical finding that Skeleton Recall Loss (SRL) exhibits an *asym-*
80 *metric* effect in knowledge distillation: SRL benefits *teacher* training (pro-
81 ducing richer representations that improve student performance by $+1.2$
82 mIoU_{fg}) but harms *student* training when combined with DTKD (-1.1
83 mIoU_{fg}), revealing that optimal training recipes differ between teachers
84 and distillation students.
- 85 3. A comprehensive evaluation including comparison with six baselines (in-
86 cluding SAM-LoRA and DINOv2-LoRA), per-tier difficulty analysis, sta-
87 tistical significance testing, and zero-shot cross-dataset transfer on S2DS,
88 demonstrating that task-specific design with knowledge distillation outper-
89 forms direct foundation-model fine-tuning on small-scale industrial datasets
90 and that DSConv’s geometric inductive bias substantially improves cross-
91 domain robustness.

92 2. Related Work

93 2.1. Concrete defect segmentation

94 Early CNN-based crack detectors such as CrackNet treated defect identi-
95 fication as patch classification (Zou et al., 2019). Fully convolutional designs

96 subsequently enabled dense pixel-wise prediction: DeepCrack (Liu et al., 2019a)
97 introduced deeply-supervised side outputs, while Zou et al. (2019) exploited hi-
98 erarchical features with multi-scale fusion.

99 Transformer-based architectures have demonstrated improved global context
100 modelling. CrackFormer (Liu et al., 2021) embeds self-attention to capture long-
101 range crack continuity, and SegFormer (Xie et al., 2021) provides a lightweight
102 hierarchical encoder with an all-MLP decoder. Universal architectures such
103 as Mask2Former (Cheng et al., 2022) further push segmentation accuracy on
104 general benchmarks but have seen limited adoption for infrastructure defects
105 due to annotation scarcity.

106 Applied to dam inspection specifically, Arafin et al. (2024) performed multi-
107 class concrete defect segmentation using DeepLabV3+, while Fang and Liu
108 (2024) proposed a lightweight architecture for microscopic dam crack quan-
109 tification. Chen et al. (2024) surveyed visual perception methods for intelligent
110 dam inspection, identifying limited dataset scale and poor topology preservation
111 as key open challenges.

112 *2.2. Topology-aware segmentation*

113 Hu et al. (2019) introduced a differentiable topological loss based on persis-
114 tent homology. Clough et al. (2020) extended persistent-homology supervision
115 to cardiac imaging. However, persistent-homology computation scales poorly to
116 high-resolution images.

117 For tubular structures, Shit et al. (2021) proposed clDice, a differentiable
118 soft-skeleton metric, but it requires GPU-based skeletonisation with substantial
119 memory overhead. The Skeleton Recall Loss (Shit et al., 2024) pre-computes
120 ground-truth skeletons on CPU and formulates a recall objective, achieving
121 superior connectivity with over 90% memory reduction relative to clDice.

122 Our work evaluates SRL in the multi-class dam defect setting, revealing
123 an asymmetric effect: SRL enriches teacher representations for distillation but
124 harms student training when combined with soft-label objectives.

125 *2.3. Deformable and dynamic convolutions*

126 Deformable Convolutional Networks (DCN) (Dai et al., 2017) introduced
127 learnable offsets for adaptive sampling. Dynamic Snake Convolution (DSConv)
128 (Qi et al., 2023) constrains offsets along a one-dimensional path, forcing snake-
129 like sampling that traces elongated structures.

130 Our DSCFormer integrates DSConv as a *parallel* branch alongside SegFormer-
131 B2, preserving global context while adding crack-specific geometric bias.

132 *2.4. Knowledge distillation for segmentation*

133 Knowledge distillation (KD) (Hinton et al., 2015) transfers knowledge from
134 a teacher model to a smaller student by matching softened output distributions.
135 In semantic segmentation, structured KD approaches (Liu et al., 2019b; Wang
136 et al., 2020) distil spatial relationships rather than just per-pixel logits. Multi-
137 teacher distillation (You et al., 2017; Yuan et al., 2021) combines knowledge
138 from multiple specialised teachers, but typically uses uniform teacher weighting
139 across all classes.

140 Our class-conditional dual-teacher strategy differs by assigning per-class
141 teacher weights based on each teacher’s domain-specific strengths: the foundation-
142 model teacher (SAM-LoRA) receives higher weight for crack classes where its
143 pretrained representations capture global connectivity, while the task-specific
144 teacher (DSCFormer) receives higher weight for spalling where domain-specific
145 features are more effective.

146 *2.5. Foundation models for segmentation*

147 The Segment Anything Model (SAM) (Kirillov et al., 2023) and DINOv2
148 (Oquab et al., 2024) represent powerful vision foundation models. Parameter-
149 efficient fine-tuning via LoRA (Hu et al., 2022) enables adaptation with minimal
150 trainable parameters ($\sim 5\%$). While promising on natural images, their effec-
151 tiveness on small-scale industrial datasets remains underexplored. We show
152 that rather than directly deploying foundation models, using them as teachers
153 in knowledge distillation yields better results.

3. Method

3.1. Overview

Fig. 1 illustrates the DSCFormer framework. Our approach enhances a SegFormer-B2 backbone at two complementary levels: (1) *feature extraction* via a parallel DSConv branch for geometry-aware crack sampling, and (2) *knowledge transfer* via class-conditional dual-teacher distillation (DTKD) to combine the strengths of task-specific and foundation models.

3.2. SegFormer-B2 backbone

We adopt SegFormer-B2 (Xie et al., 2021) as our backbone, providing a hierarchical Mix Transformer (MiT) encoder with four stages at resolutions $H/4$, $H/8$, $H/16$, and $H/32$ with channel dimensions 64, 128, 320, and 512. The all-MLP decoder fuses multi-scale features via linear projection and upsampling. We initialise from weights pretrained on ADE20K at 512×512 .

3.3. Dynamic Snake Convolution branch

Standard convolutions sample on fixed rectangular grids—poorly matched to thin, curvilinear cracks. Inspired by Qi et al. (2023), we introduce a parallel *CrackSnakeBranch* that performs deformable sampling along crack geometry.

Architecture.. The branch consumes encoder stages 1 and 2 (at $H/4$ and $H/8$ resolution). Two parallel DSConv paths deform sampling along the x -axis and y -axis respectively, each with kernel size $K=9$ and hidden dimension $d=64$. The outputs are projected to a single-channel crack enhancement map $\mathbf{E}_{\text{crack}} \in \mathbb{R}^{H \times W}$.

Additive enhancement.. The map is added *only* to the crack channel logits:

$$\hat{\mathbf{y}}_{\text{crack}} = \mathbf{z}_{\text{crack}} + \mathbf{E}_{\text{crack}} \quad (1)$$

where $\mathbf{z}_{\text{crack}}$ denotes the decoder’s crack logits. Spalling and background logits remain unmodified.

179 *Zero-initialisation..* The output projection is initialised to zero, so the branch
 180 starts as a no-op and gradually learns crack-specific deformations during train-
 181 ing.

182 3.4. *Class-conditional dual-teacher knowledge distillation*

183 A key observation from our experiments is that different model architectures
 184 excel at different defect types: the task-specific DSCFormer (Teacher 1) achieves
 185 higher spalling IoU, while the foundation-model SAM-LoRA (Teacher 2) achieves
 186 higher crack connectivity (ConnR). Standard knowledge distillation cannot ex-
 187 ploit this complementarity because it applies uniform teacher weighting across
 188 all classes.

189 We propose *class-conditional dual-teacher knowledge distillation* (DTKD)
 190 that constructs a per-class ensemble of teacher predictions:

191 *Teacher models..* We use two pretrained teachers, both trained with Skeleton
 192 Recall Loss (SRL) (Shit et al., 2024) to enrich their learned representations (the
 193 rationale for this choice is validated in Section 4.6):

- 194 • **Teacher 1** (T_1): DSCFormer (task-specific, strong on spalling)
- 195 • **Teacher 2** (T_2): SAM-LoRA (foundation model, strong on crack connec-
 196 tivity)

197 We select SAM-LoRA over DINOv2-LoRA as Teacher 2 because SAM-LoRA
 198 achieves substantially higher ConnR_{fg} (81.2% vs. 76.5%) and BF1_{fg} (68.1% vs.
 199 62.4%), providing the complementary connectivity knowledge that the task-
 200 specific teacher lacks (Table 2). DINOv2-LoRA’s advantage is limited to IoU_{sp},
 201 which Teacher 1 already covers.

202 *Class-conditional ensemble..* For each class c , the ensemble teacher probability
 203 is computed as a weighted combination:

$$\tilde{p}_c = w_{1,c} \cdot p_{1,c} + w_{2,c} \cdot p_{2,c} \quad (2)$$

204 where $p_{i,c} = \text{softmax}(\mathbf{z}_i/\tau)_c$ is the softened probability from teacher i at tem-
 205 perature τ , and $w_{1,c} + w_{2,c} = 1$ for each class.

Class-conditional weights were determined based on per-class performance analysis of the two teachers on the validation set and fixed before test-set evaluation. Since Teacher 2 (SAM-LoRA) achieves higher ConnR_{cr} while Teacher 1 achieves higher IoU_{sp} , we assign higher Teacher 2 weight for crack and higher Teacher 1 weight for spalling:

- **Background:** $w_{1,\text{bg}} = w_{2,\text{bg}} = 0.5$ (equal, neither teacher specialises)
- **Crack:** $w_{2,\text{cr}} = 0.6$ (SAM-LoRA has better crack connectivity)
- **Spalling:** $w_{2,\text{sp}} = 0.3$ (DSCFormer has better spalling coverage)

The equal-weight baseline (Table 7) shows that this assignment provides modest mIoU gains (+0.1) but notably improves crack centreline preservation (cIDice_{fg} +0.6). Manual weight selection is a limitation; adaptive class-conditional weighting is discussed in Section 5.

Distillation loss.. The ensemble probabilities are renormalised to sum to one and the student is trained with:

$$\mathcal{L}_{\text{KD}} = \tau^2 \cdot \text{KL}(\tilde{\mathbf{p}} \parallel \text{softmax}(\mathbf{z}_s/\tau)) \quad (3)$$

where $\tilde{\mathbf{p}}$ denotes the renormalised class-conditional ensemble probability and $\mathbf{z}_s$ the student logits.

Confidence-aware distillation.. When two teachers with different architectures and training recipes are combined, their predictions inevitably disagree at certain pixels—particularly at ambiguous crack-background boundaries and thin structures. At such pixels the ensemble soft label is unreliable, and forcing the student to match it may introduce noise. We quantify per-pixel teacher agreement using the KL divergence between the two softened teacher distributions:

$$d(i) = \text{KL}\left(\text{softmax}(\mathbf{z}_{T_1}^{(i)}/\tau) \parallel \text{softmax}(\mathbf{z}_{T_2}^{(i)}/\tau)\right) \quad (4)$$

where i indexes spatial positions. An agreement map is derived as $a(i) = 1 - d(i)/\max_j d(j)$, ranging from 1 (full agreement) to 0 (maximum disagreement). The confidence-aware distillation loss modulates the per-pixel KD contribution:

$$\mathcal{L}_{\text{KD}}^{\text{conf}} = \tau^2 \cdot \frac{1}{N} \sum_i a(i) \cdot \text{KL}\left(\tilde{\mathbf{p}}^{(i)} \parallel \text{softmax}(\mathbf{z}_s^{(i)}/\tau)\right) \quad (5)$$

231 This ensures the student learns strongly from the ensemble where both teach-
 232 ers concur, but falls back on the ground-truth supervised loss at ambiguous
 233 pixels. The agreement map is computed within the existing forward pass (no
 234 additional inference cost). Fig. 2 visualises the disagreement map on represen-
 235 tative Hard-tier samples: high-disagreement pixels (bright regions) concentrate
 236 at crack-background boundaries and thin branching points, precisely where the
 237 ensemble soft label is least reliable. We empirically validate that teacher dis-
 238 agreement correlates with sample difficulty: the mean per-sample KL divergence
 239 between teachers is 43% higher on Hard-tier images than on Easy-tier images
 240 (Section 4.6).

241 *Total loss.* The final training objective balances the segmentation loss and
 242 distillation loss:

$$\mathcal{L} = (1 - \alpha) [\lambda_1 \mathcal{L}_{\text{CE}} + \lambda_2 \mathcal{L}_{\text{Dice}}] + \alpha \mathcal{L}_{\text{KD}}^{\text{conf}} \quad (6)$$

243 with $\alpha = 0.5$, $\tau = 4$, and $\lambda_1 = \lambda_2 = 0.5$. Note that SRL is used only during
 244 *teacher* training; the student does not use SRL, as we find it conflicts with the
 245 distillation objective (Section 4.6).

246 3.5. Test-time augmentation

247 At inference, we optionally employ multi-scale TTA: each image is evaluated
 248 at three scales ($0.75\times$, $1.0\times$, $1.25\times$) with horizontal and vertical flips (9 views).
 249 Logits are averaged before argmax.

250 4. Experiments

251 4.1. Dataset

252 We evaluate on the **semantic segmentation subset** of the DamSegment
 253 dataset (Gharehbaghi et al., 2026), which comprises 3,500 image patches cap-
 254 tured during an on-site visit of a concrete dam, supporting classification, detec-
 255 tion, and segmentation tasks. The segmentation subset contains 1,500 pixel-level
 256 annotated images with three classes (background, crack, spalling) at 512×512

257 resolution, organised into three difficulty tiers (Easy, Medium, Hard; 500 each)
 258 based on crack detection difficulty, enabling curriculum learning. We split the
 259 data into train/val/test sets of 1,200/150/150 images, stratified by tier. Splitting
 260 is at the patch level; patches from adjacent regions may share visual character-
 261 istics (see Section 5). Fig. 3 shows representative samples from each tier with
 262 ground-truth overlays; crack pixels range from 0.3% to 7.9% and spalling from
 263 1.0% to 11.7%, illustrating the extreme class imbalance.

264 4.2. Implementation details

265 All models are trained on a single NVIDIA A40 GPU using PyTorch. DSC-
 266 Former uses SegFormer-B2 pretrained on ADE20K-512, trained for 100 epochs
 267 with AdamW (lr 6×10^{-5} , weight decay 10^{-4}), 5-epoch warmup, batch size 4.
 268 The DSConv branch uses $d=64$, $K=9$. Data augmentation consists of random
 269 horizontal and vertical flips ($p=0.5$ each); images are resized to 512×512 and
 270 normalised with ImageNet statistics.

271 For DTKD, Teacher 1 (DSCFormer+SRL) and Teacher 2 (SAM-LoRA+SRL)
 272 are trained independently first with SRL ($\lambda_3 = 0.05$, activated at epoch 60),
 273 then frozen during student training. The student has identical architecture to
 274 Teacher 1 but is trained *without* SRL (CE + Dice + KD only). Distillation uses
 275 $\alpha = 0.5$, $\tau = 4$.

276 For foundation model baselines, SAM-LoRA uses ViT-B with LoRA (rank
 277 16, $\alpha=32$) and an FPN decoder. DINOv2-LoRA adapts ViT-B/14 similarly
 278 with input size 518×518 (patch-size multiple). Both train for 50 epochs with
 279 batch size 2.

280 4.3. Evaluation metrics

281 We report complementary metrics capturing different aspects of segmenta-
 282 tion quality:

- 283 • **mIoU_{fg}**: Mean IoU of crack and spalling classes.
- 284 • **IoU_{cr}, IoU_{sp}**: Per-class IoU.
- 285 • **BF1**: Boundary F1 score (2-pixel tolerance) (Csurka et al., 2013).

Table 1: Computational cost comparison. Params: total parameters. Train: trainable parameters (all are fully trainable unless noted). FPS: images/second on a single A40 GPU (batch size 1). †: LoRA fine-tuning.

Model	Backbone	Params (M)	Train (M)	FPS
U-Net	ResNet-34	24.4	24.4	519
DeepLabV3+	ResNet-50	26.7	26.7	333
SegFormer-B2	MiT-B2	27.3	27.3	124
Mask2Former	Swin-S	68.7	68.7	39
SAM-LoRA [†]	ViT-B	93.7	5.0	—
DINOv2-LoRA [†]	ViT-B/14	92.7	6.1	68
DSCFormer	MiT-B2	31.3	31.3	96
DSCFormer+DTKD	MiT-B2	31.3	31.3	96

- **clDice**: Centreline Dice via skeletonisation (Shit et al., 2021).
- **ConnR**: Connectivity Ratio—fraction of GT connected components with $\geq 50\%$ overlap in prediction.

4.4. Computational cost

Table 1 summarises the model complexity and inference speed measured on a single NVIDIA A40 GPU at each model’s native input resolution. DSCFormer adds 4.0M parameters (+14.5%) over SegFormer-B2 due to the DSConv branch, with a modest FPS reduction (124→96). Crucially, DTKD incurs *no additional inference cost*: the two teachers are only used during training, and the student has identical architecture to the base DSCFormer. Mask2Former is the heaviest model (68.7M, 39 FPS), while foundation model baselines (SAM-LoRA, DINOv2-LoRA) carry >90M parameters due to their ViT-B encoders, though only ~5–6M are trainable via LoRA.

4.5. Comparison with state-of-the-art methods

Table 2 presents quantitative results on the DamSegment test set.

Table 2: Quantitative comparison on the DamSegment test set. Best in **bold**, second best underlined. †: LoRA fine-tuning ($\sim 5\%$ trainable parameters). ‡: trained at 320×320 . §DSCFormer is trained with SRL and serves as Teacher 1; DSCFormer+DTKD is the student trained *without* SRL (see Section 4.6). *TTA (multi-scale + flip, 9 views) applied only to our method for reference; primary comparison is without TTA. All values in %.

Method	Backbone	mIoU _{fg}	IoU _{cr}	IoU _{sp}	BF1 _{cr}	BF1 _{sp}	BF1 _{fg}	clDice _{cr}
U-Net [†] (Ronneberger et al., 2015)	ResNet-34	59.6	51.3	67.9	75.4	53.8	64.6	78.5
DeepLabV3+ (Chen et al., 2018)	ResNet-50	63.4	53.3	73.5	69.0	49.5	59.2	78.5
Mask2Former (Cheng et al., 2022)	Swin-S	67.9	57.1	78.6	76.0	69.0	72.5	81.2
SegFormer-B2 (Xie et al., 2021)	MiT-B2	70.4	56.9	83.8	73.7	59.8	66.8	81.2
SAM-LoRA [†] (Kirillov et al., 2023)	ViT-B	68.1	55.9	80.2	73.1	63.0	68.1	81.2
DINOv2-LoRA [†] (Oquab et al., 2024)	ViT-B	66.7	49.1	84.2	64.7	60.1	62.4	77.5
DSCFormer [§]	MiT-B2	70.6	57.3	83.8	74.7	66.0	70.3	83.6
DSCFormer + DTKD	MiT-B2	<u>72.3</u>	<u>58.5</u>	<u>86.2</u>	<u>75.6</u>	65.6	70.6	83.6
DSCFormer + DTKD + TTA*	MiT-B2	72.7	59.3	86.0	76.3	<u>67.2</u>	<u>71.8</u>	83.6

301 DSCFormer with DTKD achieves the highest mIoU_{fg} (72.3% without TTA,
302 72.7% with TTA) and the highest IoU for both crack and spalling classes
303 among all compared methods. Mask2Former achieves the highest BF1_{fg} (72.5%)
304 due to its mask-attention mechanism, though with substantially lower mIoU_{fg}
305 (67.9%). DSCFormer+DTKD+TTA surpasses Mask2Former on BF1_{cr} (76.3%
306 vs. 76.0%). DSCFormer+DTKD achieves ConnR_{fg} 83.0%, surpassing SAM-
307 LoRA (81.2%); this connectivity primarily reflects DSConv’s geometric induc-
308 tive bias (the DSConv-only ablation variant achieves 83.6%, with DTKD trading
309 -0.6 ConnR_{fg} for its pixel-level gains). Compared to the SegFormer-B2 base-
310 line, DSCFormer+DTKD improves mIoU_{fg} by **+1.9**, IoU_{cr} by **+1.6**, IoU_{sp} by
311 **+2.4**, and ConnR_{fg} by **+2.1**.

312 *Benefit of DTKD.* In the ablation (Table 3, row b→c), adding DTKD to the
313 DSConv-only base improves region-level metrics substantially: mIoU_{fg} +1.9,
314 IoU_{cr} +1.6, IoU_{sp} +2.2, and clDice_{cr} +1.3, at a modest cost to crack connec-

315 tivity ($\text{ConnR}_{\text{cr}} -2.6$, net $\text{ConnR}_{\text{fg}} -0.6$). The resulting ConnR_{fg} (83.0%) still
 316 surpasses SAM-LoRA (81.2%), confirming that the DSConv geometric prior,
 317 partially attenuated by DTKD’s soft-label supervision, retains strong connec-
 318 tivity.

319 *Foundation model comparison..* Despite leveraging powerful pretrained repre-
 320 sentations, SAM-LoRA (68.1%) and DINOv2-LoRA (66.7%) substantially un-
 321 derperform both the base DSCFormer (70.6%) and DSCFormer+DTKD (72.3%).
 322 However, our DTKD framework demonstrates that foundation model knowledge
 323 can be *indirectly* leveraged: rather than deploying SAM directly, distilling its
 324 knowledge into the student yields ConnR_{fg} 83.0% (primarily from DSConv; see
 325 ablation), surpassing SAM-LoRA’s own 81.2%. This represents a more effective
 326 use of foundation models on small-scale industrial datasets.

327 Mask2Former also achieves the highest $\text{clDice}_{\text{sp}}$ (91.1%), but with substan-
 328 tially lower mIoU_{fg} (67.9%) and IoU_{sp} (78.6%), indicating poor spalling coverage
 329 despite good boundary and shape capture.

330 4.6. Ablation study

331 Table 3 presents the incremental contribution of each component.

332 *(a)→(b): Adding DSConv..* The DSConv branch produces minimal mIoU change
 333 (+0.1) but substantially improves crack connectivity: ConnR_{cr} increases by
 334 +**3.7** (81.0→84.7) and $\text{clDice}_{\text{fg}}$ by +**2.8** (82.7→85.5). The deformable snake-
 335 shaped sampling captures elongated crack geometry that fixed rectangular ker-
 336 nels miss, demonstrating that DSConv’s primary value lies in its geometric
 337 inductive bias for connectivity.

338 *(b)→(c): Adding DTKD..* DTKD provides substantial pixel-level improvement:
 339 $\text{mIoU}_{\text{fg}} +1.9$, $\text{IoU}_{\text{cr}} +1.6$, $\text{IoU}_{\text{sp}} +2.2$, $\text{clDice}_{\text{cr}} +1.3$, and $\text{BF1}_{\text{cr}} +1.9$. However,
 340 ConnR_{cr} decreases from 84.7 to 82.1 (−2.6), as DTKD’s soft-label supervi-
 341 sion slightly smooths fine crack structures; this is partially offset by improved
 342 spalling connectivity, yielding a net ConnR_{fg} change of only −0.6 (83.6→83.0).

Table 3: Ablation study on the DamSegment test set. Rows (a)–(d) show the main pipeline progression; row (g) adds confidence-aware KD to the best model; rows (e)–(f) show the effect of adding SRL. Row (a) retrains SegFormer-B2 using the same training pipeline as all ablation variants. All values in %.

Configuration	mIoU _{fg}	IoU _{cr}	IoU _{sp}	BF1 _{cr}	BF1 _{fg}	clDice _{cr}	clDice _{fg}	ConnR _{cr}	ConnR _{fg}
(a) SegFormer-B2	70.3	56.9	83.7	74.0	68.6	81.8	82.7	81.0	77.0
(b) + DSConv	70.4	56.9	84.0	73.7	69.5	82.0	85.5	84.7	83.6
(c) + DTKD	72.3	58.5	86.2	75.6	70.6	83.3	86.4	82.1	83.0
(g) + conf-aware KD	<u>72.5</u>	<u>58.7</u>	<u>86.3</u>	<u>76.4</u>	<u>71.1</u>	<u>83.9</u>	<u>86.1</u>	81.4	<u>82.6</u>
(d) (c) + TTA	72.7	59.3	86.0	76.3	71.8	83.7	86.2	79.4	80.2
(e) (b) + SRL	70.6	57.3	83.8	74.7	70.3	82.5	85.7	82.2	77.8
(f) (e) + DTKD	71.2	58.0	84.5	75.4	71.4	83.2	86.2	82.3	80.6

343 The class-conditional weighting ensures that the student receives the most in-
344 formative teacher guidance for each defect type (Section 3.4).

345 *(c)→(g): Confidence-aware KD.* Replacing the standard KD loss with confidence-
346 aware distillation (Eq. 5) improves boundary quality: BF1_{cr} +0.8 (75.6→76.4)
347 and clDice_{cr} +0.6 (83.3→83.9), with a small mIoU_{fg} gain (+0.2). This confirms
348 that down-weighting the distillation signal at high-disagreement pixels reduces
349 noise from unreliable ensemble labels. ConnR_{cr} decreases modestly (82.1→81.4),
350 as the reduced KD weight at ambiguous boundary pixels allows the student to
351 rely more on ground truth—which improves pixel precision but slightly reduces
352 connectivity.

353 To isolate the contribution of confidence-aware KD from difficulty-guided
354 sample reweighting (an alternative use of teacher disagreement), we conducted
355 an ablation using disagreement-based per-sample loss reweighting *without* pixel-
356 level confidence modulation. This variant degraded performance (mIoU_{fg} 72.1,
357 BF1_{fg} 70.0), indicating that on small datasets, per-sample reweighting intro-
358 duces training instability, while pixel-level modulation of the distillation signal
359 is the effective mechanism.

Analysis of teacher disagreement across difficulty tiers reveals that the mean per-sample KL divergence between teachers is 0.046 on Hard-tier images vs. 0.032 on Easy-tier images (+43%), confirming that disagreement is an informative proxy for sample difficulty. However, this correlation does not translate into effective sample-level curriculum learning—consistent with prior findings that data-restricting or reweighting curriculum strategies underperform on small datasets where every sample matters.

(c)→(d): Test-time augmentation.. TTA provides modest additional gains (mIoU_{fg} +0.4, IoU_{cr} +0.8) but reduces connectivity metrics (ConnR_{fg} 83.0→80.2), suggesting that multi-scale/flip aggregation improves pixel coverage but may smooth fine structural details.

Effect of SRL (rows e–f).. Skeleton Recall Loss (SRL) (Shit et al., 2024) penalises low predicted probability at ground-truth skeleton pixels $\mathcal{S}$, precomputed offline via morphological skeletonisation:

$$\mathcal{L}_{\text{SRL}} = 1 - \frac{1}{|\mathcal{S}|} \sum_{p \in \mathcal{S}} \hat{y}_{\text{crack}}(p) \quad (7)$$

We apply SRL only to the crack class (activated at epoch 60, weight $\lambda_3 = 0.05$). Rows (e)–(f) show the effect of adding SRL to the pipeline. Adding SRL to DSConv (row e) provides small crack improvements (IoU_{cr} +0.4, BF1_{cr} +1.0) but severely damages connectivity: ConnR_{fg} drops from 83.6 to 77.8 (−5.8), driven by ConnR_{sp} falling from 82.4 to 73.4 as SRL’s crack-focused supervision fragments spalling components at ambiguous boundaries. Adding DTKD on top of SRL (row f) partially recovers connectivity (77.8→80.6) but the resulting model (f, mIoU_{fg} 71.2%) underperforms the SRL-free pipeline (c, mIoU_{fg} 72.3%) by **−1.1**. This demonstrates that SRL’s gradient interference with DTKD’s soft-label supervision is a net negative: the student learns better from teacher signals alone without the competing skeleton objective. Multi-seed experiments (Table 4) further confirm that the SRL path is unstable (mIoU_{fg} std 2.42 vs. 0.52 for DTKD alone).

387 *Overall progression..* From baseline (a) to the full method with confidence-
388 aware KD (g), the improvements are: $\text{mIoU}_{\text{fg}} +2.2$, $\text{IoU}_{\text{cr}} +1.8$, $\text{IoU}_{\text{sp}} +2.6$,
389 $\text{BF1}_{\text{fg}} +2.5$, $\text{clDice}_{\text{fg}} +3.4$, and $\text{ConnR}_{\text{fg}} +5.6$. The three components are
390 complementary: DSConv provides the geometric inductive bias for connectivity
391 ($\text{ConnR}_{\text{fg}} +6.6$), DTKD adds pixel-level accuracy across all metrics at a modest
392 cost to crack connectivity ($\text{ConnR}_{\text{cr}} -2.6$, net $\text{ConnR}_{\text{fg}} -0.6$), and confidence-
393 aware KD further refines boundary quality ($\text{BF1}_{\text{cr}} +0.8$) by suppressing noisy
394 distillation at ambiguous pixels. DTKD also acts as a strong regulariser: multi-
395 seed experiments (Table 4) show it reduces cross-seed variance by $\sim 3\times$ (mIoU_{fg}
396 std: 0.52 vs. 1.47 for the baseline), confirming that distillation is the critical
397 component for both performance and reliability.

398 *Statistical significance..* To verify that DTKD’s gains are not due to random
399 variation, we conduct paired Wilcoxon signed-rank tests on the test images,
400 comparing DSConv-only (row b) vs. DSConv+DTKD (row c, final model).
401 Six of ten per-class metrics reach significance at $\alpha=0.05$: IoU_{cr} ($p=0.008$),
402 Dice_{cr} ($p=0.007$), BF1_{cr} ($p=0.046$), $\text{clDice}_{\text{cr}}$ ($p<0.001$), IoU_{sp} ($p=0.016$), and
403 Dice_{sp} ($p=0.016$). The four non-significant metrics (BF1_{sp} , $\text{clDice}_{\text{sp}}$, ConnR_{cr} ,
404 ConnR_{sp}) all involve small paired-sample sizes ($n=31\text{--}37$) and show direction-
405 ally consistent improvements, except ConnR_{cr} where DSConv-only is marginally
406 better (-1.7 , $p=0.040$). Tier-stratified analysis reveals the significance concen-
407 trates on Hard images: IoU_{cr} ($p=0.003$), Dice_{cr} ($p=0.004$), BF1_{cr} ($p=0.045$),
408 and $\text{clDice}_{\text{cr}}$ ($p<0.001$) all reach significance, confirming that the foundation-
409 model teacher’s global knowledge provides the strongest benefit on complex
410 crack patterns.

411 *Multi-seed training stability..* To assess robustness to random initialisation, we
412 train three key configurations—SegFormer-B2 (baseline), DSCFormer+SRL,
413 and DSCFormer+DTKD (row c, final model)—with three independent seeds
414 (42, 123, 2024) and report mean $\pm$ standard deviation in Table 4.

415 DTKD achieves the highest mean performance on all six metrics while ex-
416 hibiting the lowest variance on four of six (std ≤ 0.85 ; IoU_{sp} and ConnR_{fg} show

Table 4: Multi-seed stability analysis (mean $\pm$ std over 3 seeds). DSCFormer+DTKD is the final model (row c, no SRL). All values in %.

Metric	SegFormer-B2	DSCFormer+SRL	DSCFormer+DTKD
mIoU _{fg}	69.2 $\pm$ 1.47	67.9 $\pm$ 2.42	71.8 $\pm$ 0.52
IoU _{cr}	55.5 $\pm$ 2.08	53.7 $\pm$ 3.89	58.5 $\pm$ 0.31
IoU _{sp}	82.9 $\pm$ 0.86	82.2 $\pm$ 0.95	85.1 $\pm$ 1.07
BF1 _{fg}	67.1 $\pm$ 1.62	68.5 $\pm$ 2.84	71.1 $\pm$ 0.85
clDice _{fg}	81.8 $\pm$ 1.93	82.1 $\pm$ 3.44	86.3 $\pm$ 0.34
ConnR _{fg}	80.3 $\pm$ 2.13	75.7 $\pm$ 3.68	81.3 $\pm$ 1.43

slightly higher variance due to spalling’s inherent variability across seeds). Notably, DSCFormer+SRL shows the highest variance (mIoU_{fg} std = 2.42), and its multi-seed mean (67.9%) is *lower* than the baseline SegFormer-B2 (69.2%), indicating that the single-seed improvement reported in Table 3 is not robust. This underscores DTKD’s dual role: it not only improves absolute performance but also acts as a strong regulariser, reducing sensitivity to initialisation by $\sim 3\times$ compared to the baseline (mIoU_{fg} std: 0.52 vs. 1.47) and $\sim 5\times$ compared to DSCFormer+SRL (0.52 vs. 2.42). The stabilisation effect likely stems from the soft-label supervision provided by two pretrained teachers, which smooths the loss landscape and reduces the impact of random initialisation on convergence.

Per-tier breakdown.. Table 5 reports per-image mean performance stratified by difficulty tier, comparing DSConv-only (row b) and DSConv+DTKD (row c, final model) to isolate DTKD’s contribution. DTKD improves mIoU_{fg} consistently across all tiers (+3.4 Easy, +2.3 Medium, +2.4 Hard), but the gains reach statistical significance only on Hard images (IoU_{cr} $p=0.003$, clDice_{cr} $p<0.001$), where complex crack patterns benefit most from the foundation-model teacher. The spalling improvements are largest on Easy (+6.2 IoU_{sp}, $p=0.035$) where the SAM-LoRA teacher’s global context helps distinguish large spalling regions. ConnR_{fg} improves on Easy (+4.8) and Hard (+1.4) tiers, driven by ConnR_{sp} gains, while ConnR_{cr} shows a small decrease on Hard (−2.4, $p=0.018$)—a trade-

Table 5: Per-tier performance breakdown (per-image means). Bold: better value per tier. Wilcoxon p -values reported in text. All values in %.

Tier	Model	mIoU _{fg}	IoU _{cr}	IoU _{sp}	BF1 _{fg}	clDice _{fg}	ConnR _{fg}
Easy	DSCnv (b)	62.2	59.8	64.6	66.8	86.7	84.2
	DSCnv+DTKD (c)	65.6	60.4	70.8	68.9	88.8	89.0
Medium	DSCnv (b)	56.2	59.7	52.7	77.4	87.3	74.8
	DSCnv+DTKD (c)	58.5	60.8	56.2	78.4	90.5	74.2
Hard	DSCnv (b)	61.9	52.8	71.0	72.6	84.1	81.7
	DSCnv+DTKD (c)	64.2	53.8	74.6	74.1	85.5	83.1

Table 6: Individual teacher performance vs. DTKD student on the test set. The student surpasses both teachers, confirming effective knowledge combination. All values in %.

Model	mIoU _{fg}	IoU _{cr}	IoU _{sp}	ConnR _{cr}	ConnR _{sp}
Teacher 1 (DSCFormer)	70.6	57.3	83.8	82.2	73.4
Teacher 2 (SAM-LoRA+SRL)	69.0	55.9	82.1	82.2	86.0
Student (DTKD)	72.3	58.5	86.2	82.1	<u>83.8</u>

off where DTKD slightly sacrifices crack connectivity for substantially better
spalling connectivity and overall accuracy.

4.7. Analysis of DTKD teacher contributions

Table 6 compares individual teachers with the DTKD student.

The DTKD student surpasses both teachers on mIoU_{fg} (+1.7 over Teacher 1, +3.3 over Teacher 2), IoU_{cr}, and IoU_{sp}, confirming that class-conditional distillation effectively combines complementary strengths. Teacher 2 (SAM-LoRA+SRL) achieves the highest ConnR_{sp} (86.0%), reflecting SAM’s strong global structure preservation. The student substantially inherits this advantage (ConnR_{sp} = 83.8%, vs. Teacher 1’s 73.4%), while maintaining superior pixel-level accuracy. However, the student does not fully match Teacher 2’s ConnR_{sp} (83.8% vs.

Table 7: Effect of class-conditional teacher weighting in DTKD (final model, no SRL). “Equal” uses $w_{T2}=0.5$ for all classes; “Class-cond.” uses $w_{T2}=0.6$ for crack, $w_{T2}=0.3$ for spalling. Bold: better value per metric. All values in %.

Weighting	mIoU _{fg}	IoU _{cr}	IoU _{sp}	BF1 _{fg}	clDice _{fg}	ConnR _{fg}
Equal (0.5/0.5)	72.2	58.4	86.0	71.5	85.8	83.1
Class-cond.	72.3	58.5	86.2	70.6	86.4	83.0

86.0%) and slightly trails both teachers on ConnR_{cr} (82.1% vs. 82.2%), consistent with the ConnR_{cr} trade-off observed in the ablation.

The student surpassing both teachers is consistent with multi-teacher distillation findings in the literature (You et al., 2017; Yuan et al., 2021), where combining complementary knowledge sources provides regularisation benefits beyond what any single teacher offers. Unlike classical “born-again” networks (Furlanello et al., 2018) where student and teacher share the same architecture, our student additionally benefits from the DSConv geometric inductive bias that neither teacher possesses.

Class-conditional weighting ablation.. Table 7 compares equal-weight (0.5/0.5) and class-conditional teacher weighting in the final SRL-free pipeline. The two configurations achieve nearly identical mIoU_{fg} (72.3 vs. 72.2), but class-conditional weighting improves crack centreline preservation: clDice_{fg} +0.6 (85.8→86.4), at the cost of slightly lower BF1_{fg} (−0.9). The result suggests that class-conditional weighting primarily steers the distillation signal toward better crack structure preservation rather than improving overall pixel accuracy, consistent with the design intent of assigning higher SAM-LoRA weight for crack connectivity.

Teacher selection ablation.. Since the DTKD student does not use SRL, one might expect a teacher trained *without* SRL (higher ConnR_{fg}) to provide cleaner knowledge transfer. We test this by replacing Teacher 1 (DSCFormer+SRL, ConnR_{fg} 77.8%) with the SRL-free DSConv-only variant (ConnR_{fg} 83.6%).

470 The resulting student achieves 71.1% mIoU_{fg} , -1.2 below the G1 teacher vari-
 471 ant (72.3%). The degradation concentrates on spalling ($\text{IoU}_{\text{sp}} -2.4$, ConnR_{sp}
 472 -1.4), while crack metrics are largely unaffected. This indicates that SRL-
 473 trained Teacher 1 learns richer internal representations—despite its own con-
 474 nectivity regression—that transfer more effectively through soft-label distilla-
 475 tion. Teacher quality for KD is not simply a function of the teacher’s own test
 476 metrics; the diversity of learned features matters more.

477 4.8. Qualitative results

478 Fig. 4 presents representative predictions across all three difficulty tiers, com-
 479 paring SegFormer-B2, DSCFormer, and DSCFormer+DTKD. SegFormer-B2
 480 produces severe over-segmentation (especially on Easy and Hard samples), while
 481 DSCFormer substantially improves crack localisation. DSCFormer+DTKD fur-
 482 ther refines both crack connectivity and spalling coverage, with per-image IoU
 483 scores consistently higher than DSCFormer alone.

484 4.9. Cross-dataset evaluation

485 To assess generalisability beyond the training domain, we perform zero-shot
 486 transfer evaluation on the S2DS dataset (Benz and Rodehorst, 2022), which con-
 487 tains 743 images (1024×1024) of concrete surfaces with pixel-level annotations
 488 for seven defect classes: background, crack, spalling, corrosion, efflorescence,
 489 vegetation, and control point. We map S2DS crack and spalling to our label
 490 scheme and treat all other classes as background, since they fall outside the
 491 DamSegment label space. All images are resized to 512×512 (bilinear for im-
 492 ages, nearest-neighbour for masks). The S2DS test split contains 93 images, of
 493 which 27 contain only crack, 20 contain only spalling, and 46 are background-
 494 only; crack pixels account for 0.12% and spalling for 4.27% of the test area. No
 495 fine-tuning is performed: models trained on DamSegment are applied directly.

496 Table 8 reveals a large domain gap for all models. SegFormer-B2 and
 497 Mask2Former fail almost completely on the unseen domain ($\text{mIoU}_{\text{fg}} < 2.5\%$),
 498 indicating severe overfitting to the DamSegment distribution. Models equipped

Table 8: Cross-dataset evaluation: DamSegment→S2DS (zero-shot). [§]DSCFormer is the SRL variant (Teacher 1). All models evaluated without TTA. *TTA row shown for reference only. All values in %.

Model	mIoU _{fg}	IoU _{cr}	IoU _{sp}	BF1 _{fg}	clDice _{fg}	ConnR _{fg}
SegFormer-B2	0.1	0.1	0.0	2.8	1.3	49.9
Mask2Former	2.5	0.1	4.8	5.5	21.2	49.7
DSCFormer [§]	17.1	7.6	26.7	49.6	72.1	69.3
DSCFormer + DTKD	18.7	8.6	28.9	46.6	68.4	70.1
DSCFormer + DTKD + TTA*	18.6	8.4	28.9	45.3	69.0	69.0

with DSConv achieve substantially higher relative transfer performance (mIoU_{fg} >17%), though the absolute values remain limited, reflecting the inherent difficulty of zero-shot cross-domain segmentation under significant distribution shift. DSCFormer+DTKD achieves the highest mIoU_{fg} (18.7%; TTA does not improve this, yielding 18.6%), while DSCFormer (Teacher 1, with SRL) achieves the highest BF1_{fg} (49.6%) and clDice_{fg} (72.1%).

These results suggest that DSConv’s geometric inductive bias improves cross-domain robustness, as crack and spalling morphology shares common geometric patterns across datasets. DTKD further improves cross-domain transfer (+1.6 mIoU_{fg} over DSCFormer alone), while also maximising in-domain accuracy (+1.9 mIoU_{fg} on DamSegment). Unlike in-domain evaluation, TTA does not improve cross-dataset performance (18.7%→18.6%), suggesting that multi-scale aggregation offers diminishing returns under large domain shift.

Fig. 5 shows representative S2DS predictions. SegFormer-B2 and Mask2Former produce largely incorrect segmentations, while DSCFormer and DSCFormer+DTKD localise defect regions despite the domain shift, consistent with the quantitative results above.

516 5. Discussion

517 *DTKD as the critical component.* A key finding is that DTKD alone—without
 518 SRL—provides the strongest and most reliable improvement. Adding DTKD
 519 to DSConv yields $+1.9$ mIoU_{fg} and $+2.4$ IoU_{sp}, though at a modest cost to
 520 crack connectivity: ConnR_{cr} drops from 84.7 to 82.1 (-2.6), as the soft-label
 521 supervision slightly smooths fine crack structures. This is partially offset by
 522 improved spalling connectivity, yielding a net ConnR_{fg} change of only -0.6
 523 ($83.6 \rightarrow 83.0$). The trade-off reflects that pixel-level and topological objectives
 524 are not perfectly aligned; practitioners prioritising crack path continuity over
 525 overall pixel accuracy may prefer the DSConv-only variant. In contrast, the
 526 SRL+DTKD path (Table 3, row f) achieves only 71.2% mIoU_{fg}—1.1 points
 527 lower—because SRL’s skeleton-level gradients conflict with DTKD’s soft-label
 528 supervision. The class-conditional weighting steers the distillation signal toward
 529 better crack centreline preservation (cIDice_{fg} $+0.6$ over equal weighting) by as-
 530 signing higher SAM-LoRA weight for crack (0.6) and higher DSCFormer weight
 531 for spalling (0.7), though the overall mIoU difference is marginal ($+0.1$). Beyond
 532 metric improvement, DTKD serves as a powerful training regulariser: multi-seed
 533 experiments (Table 4) show that DTKD reduces cross-seed variance by $\sim 3\text{--}5\times$
 534 (mIoU_{fg} std: 0.52 vs. 1.47 for the baseline, vs. 2.42 for DSCFormer+SRL). The
 535 soft-label supervision from two pretrained teachers smooths the loss landscape,
 536 making convergence less sensitive to random initialisation—a practical advan-
 537 tage for deployment scenarios where retraining reliability matters.

538 *Teacher disagreement as a quality signal.* An advantage of multi-teacher distil-
 539 lation is that the *disagreement* between teachers provides a free quality signal.
 540 We find that per-pixel KL divergence between teachers is 43% higher on Hard-
 541 tier images than Easy-tier images, confirming that disagreement correlates with
 542 sample difficulty. Leveraging this signal at the pixel level (confidence-aware
 543 KD) is effective: down-weighting distillation at high-disagreement pixels im-
 544 proves BF1_{cr} by $+0.8$ and cIDice_{cr} by $+0.6$. However, using the same sig-
 545 nal for *sample-level* curriculum learning (reweighting entire samples by their

546 aggregate disagreement) degrades performance, consistent with prior findings
 547 that curriculum strategies that alter per-sample training intensity are counter-
 548 productive on small datasets where every sample is needed for generalisation.
 549 This distinction—pixel-level modulation helps, sample-level modulation hurts—
 550 suggests that the granularity of the intervention matters more than the quality
 551 of the difficulty signal.

552 *SRL: useful for teachers, harmful for students..* An intriguing finding is that
 553 SRL plays opposite roles depending on where it is applied. As a *teacher training*
 554 component, SRL is beneficial: Teacher 1 (trained with SRL) produces a better
 555 DTKD student (72.3%) than a teacher trained without SRL (71.1%), despite the
 556 SRL teacher’s own lower ConnR_{fg} (77.8% vs. 83.6%). This suggests SRL forces
 557 the teacher to learn richer internal representations that transfer well through
 558 distillation. However, as a *student training* component, SRL is harmful: it
 559 interferes with DTKD’s soft-label supervision, fragmenting spalling predictions
 560 and increasing training instability. This asymmetry highlights that the optimal
 561 training recipe for a teacher may differ from that of a distillation student.

562 *Indirect use of foundation models..* Our results reveal a practical strategy for
 563 leveraging foundation models on small industrial datasets: rather than directly
 564 fine-tuning SAM or DINOv2 (which underperforms task-specific architectures
 565 by 2–4% mIoU), using them as *teachers* in knowledge distillation allows the
 566 student to inherit their structural knowledge while maintaining the task-specific
 567 model’s advantages. This is more parameter-efficient at inference time (student
 568 has identical cost to the base model) and more effective than direct deployment.

569 *DSCConv mechanism..* The DSCConv branch’s effectiveness stems from its geo-
 570 metric inductive bias: by constraining kernel deformations to follow elongated
 571 paths, it captures crack features that rectangular convolutions systematically
 572 miss. The zero-initialisation ensures the branch only activates when beneficial.
 573 The ConnR improvement (+6.6 in ablation) confirms that DSCConv preserves
 574 crack connectivity at the connected-component level.

575 *Practical implications..* From an inspection workflow perspective, DSCFormer+DTKD
 576 offers a practical balance: the student model has identical inference cost to the
 577 base DSCFormer (96 FPS, 31.3M parameters), with the two teachers used only
 578 during training. The per-tier analysis suggests prioritising model deployment on
 579 Hard-tier regions where DTKD provides the largest benefit, while Easy regions
 580 may not require topology-aware enhancements. For integration with existing
 581 dam SHM pipelines, the model’s crack connectivity (ConnR_{fg} 83.0%, driven
 582 primarily by DSConv) helps produce predictions that are directly usable for
 583 crack length and density estimation without requiring extensive post-processing
 584 to reconnect fragmented segments.

585 *Failure modes..* The model underperforms in two scenarios: (i) thin hairline
 586 cracks (<2 pixels wide) where even DSConv’s deformable sampling cannot cap-
 587 ture sufficient signal, and (ii) regions where crack and spalling co-occur, where
 588 the model tends to over-predict one class at the expense of the other.

589 *Limitations..* The DamSegment dataset comprises images from a single dam
 590 site, and the train/val/test split is at the patch level; spatially adjacent patches
 591 may share visual characteristics, potentially inflating in-domain metrics. Scene-
 592 level or location-based splitting would provide a more conservative evaluation.
 593 The S2DS cross-dataset transfer partially mitigates this concern but reveals
 594 a substantial domain gap ($\sim 19\%$ mIoU_{fg}), indicating that in-domain results
 595 should not be directly extrapolated to other sites. Multi-site datasets would
 596 strengthen generalisability claims. The per-class teacher weights in DTKD
 597 are set manually based on validation-set performance analysis; adaptive class-
 598 conditional weighting is an avenue for future work. Crack IoU remains challeng-
 599 ing ($\sim 59\%$) due to extreme class imbalance (crack pixels $<5\%$ of total area).
 600 The DSConv branch is crack-specific; extending to other elongated defects is
 601 future work. Cross-dataset evaluation on S2DS demonstrates improved relative
 602 robustness of DSConv-equipped models, but the absolute mIoU_{fg} remains low
 603 ($\sim 19\%$), indicating that zero-shot transfer across concrete inspection domains is

604 far from solved; domain adaptation or mixed-dataset training may be necessary
605 for practical cross-site deployment.

606 6. Conclusion

607 We presented DSCFormer, a topology-aware segmentation framework for
608 concrete dam defect analysis incorporating two complementary enhancements
609 to a SegFormer-B2 backbone. Dynamic Snake Convolution drives crack connec-
610 tivity ($\text{ConnR}_{\text{fg}} + 6.6$) with only 4M additional parameters, and dual-teacher
611 knowledge distillation (DTKD) drives pixel-level accuracy ($\text{mIoU}_{\text{fg}} + 1.9$ over
612 DSCConv alone) at zero additional inference cost, while reducing cross-seed vari-
613 ance by $\sim 3\text{--}5\times$. Confidence-aware distillation, which modulates KD inten-
614 sity based on pixel-level teacher agreement, further improves boundary quality
615 ($\text{BF1}_{\text{cr}} + 0.8$, $\text{clDice}_{\text{cr}} + 0.6$). DTKD trades modest crack connectivity ($\text{ConnR}_{\text{cr}} - 2.6$) for substantial pixel-level gains; the net ConnR_{fg} (-0.6) reflects improved
616 spalling connectivity partially offsetting the crack regression. We also investi-
617 gated Skeleton Recall Loss and found an asymmetric effect: SRL benefits teacher
618 training (producing richer representations for distillation) but harms student
619 training when combined with DTKD ($-1.1 \text{ mIoU}_{\text{fg}}$), highlighting that optimal
620 training recipes differ between teachers and distillation students. The full frame-
621 work achieves 72.3% foreground mIoU (72.7% with TTA), the highest among all
622 compared methods on the DamSegment benchmark. ConnR_{fg} (83.0%) primar-
623 ily reflects DSCConv’s contribution; the DSCConv-only variant achieves a slightly
624 higher 83.6%. Class-conditional teacher weighting modestly improves crack cen-
625 treline preservation ($\text{clDice}_{\text{fg}} + 0.6$) over equal weighting, serving as a prac-
626 tical design choice within DTKD rather than a core contribution. Zero-shot
627 cross-dataset evaluation on S2DS confirms that DSCConv’s geometric inductive
628 bias substantially improves cross-domain robustness (18.7% vs. 0.1% mIoU_{fg}),
629 though absolute performance remains limited, motivating future work on do-
630 main adaptation. Our work demonstrates that indirectly leveraging foundation
631 models through knowledge distillation is more effective than direct parameter-
632

633 efficient fine-tuning on small-scale industrial datasets, providing practical guid-
634 ance for deploying deep learning in data-scarce inspection scenarios.

635 **Data availability**

636 The DamSegment dataset is publicly available (Gharehbaghi et al., 2026).
637 Source code will be made available upon publication.

638 **Declaration of competing interest**

639 The authors declare no competing interests.

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641 [To be added.]

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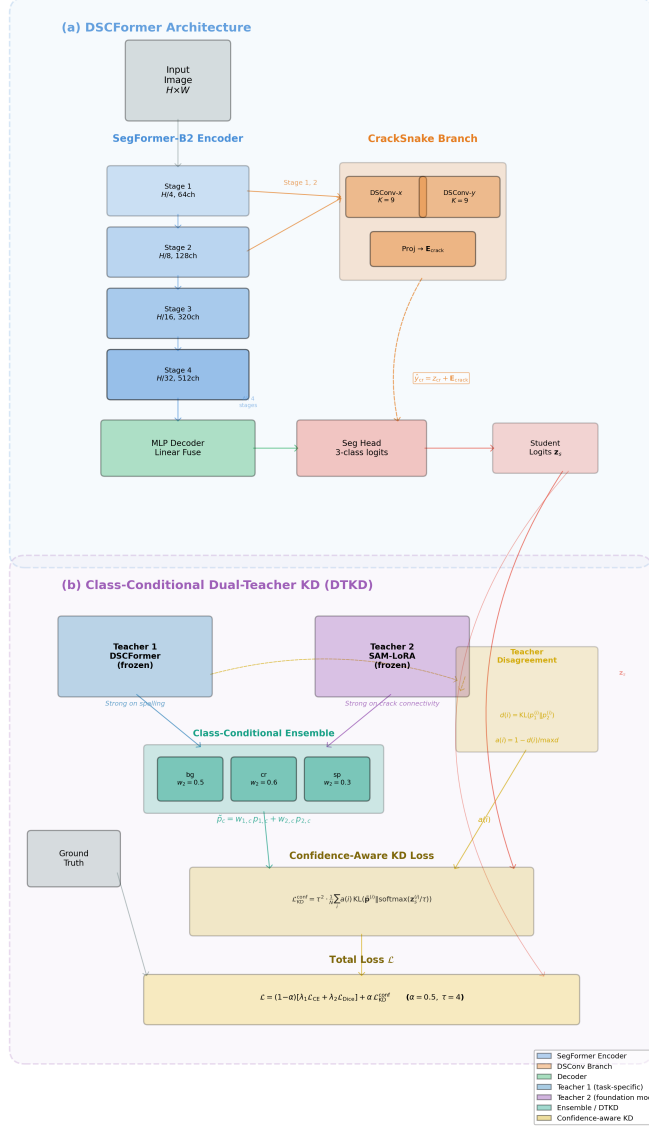


Figure 1: Overview of the DSCFormer framework. (a) A SegFormer-B2 encoder with a parallel CrackSnake branch (DSConv) produces 3-class segmentation logits; the DSConv output additively enhances only the crack channel. (b) Class-conditional dual-teacher knowledge distillation (DTKD): Teacher 1 (DSCFormer, strong on spalling) and Teacher 2 (SAM-LoRA, strong on crack connectivity) are combined via per-class weights into an ensemble. A pixel-level agreement map $a(i)$, derived from teacher disagreement, modulates the KD loss so the student learns strongly where teachers agree and falls back on ground-truth supervision at ambiguous pixels.

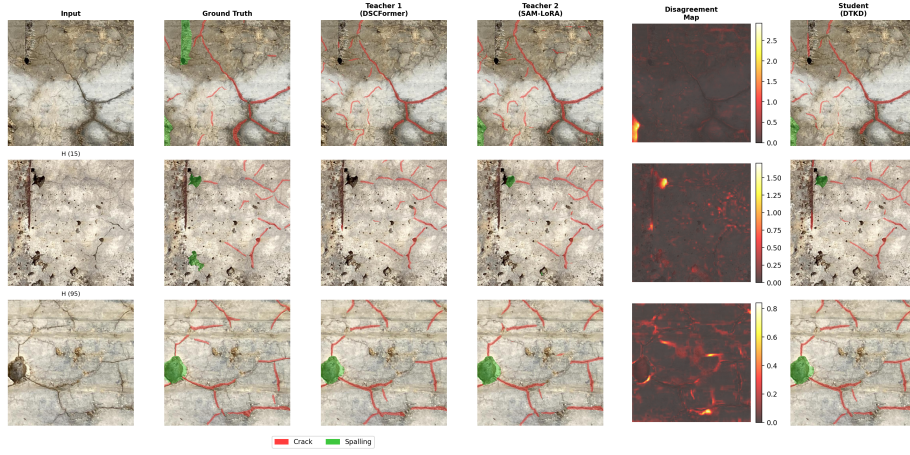


Figure 2: Teacher disagreement visualisation on Hard-tier samples. Columns: input, ground truth, Teacher 1 (DSCFormer) prediction, Teacher 2 (SAM-LoRA) prediction, pixel-level KL divergence heatmap (bright = high disagreement), student (DTKD) prediction. Disagreement concentrates at crack boundaries and thin structures where the ensemble soft label is unreliable. Red: crack, green: spalling.

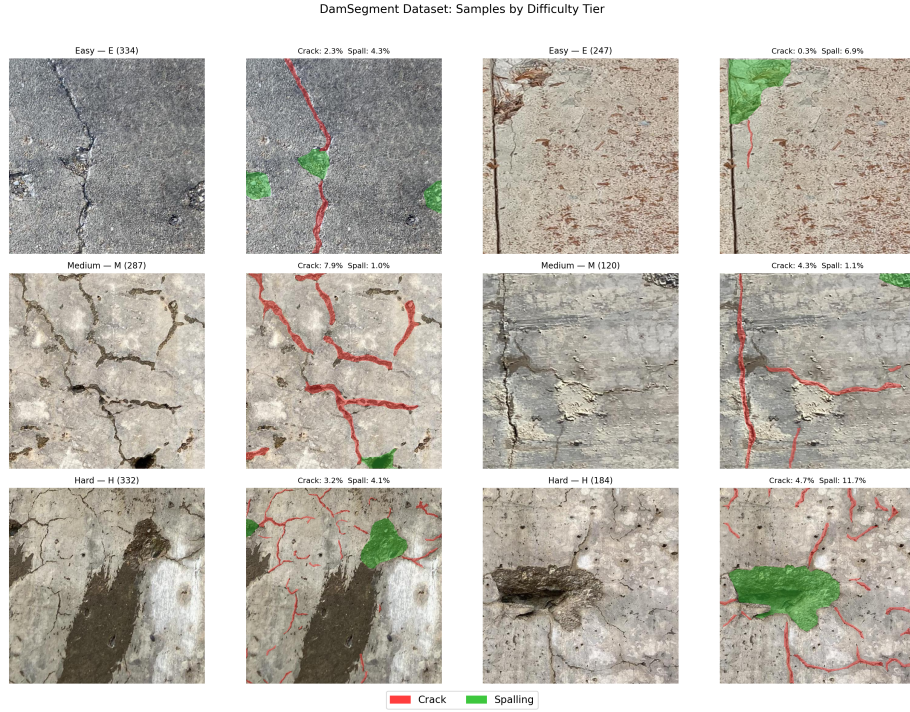


Figure 3: DamSegment dataset samples by difficulty tier. Each pair shows the raw image (left) and ground-truth overlay (right). Red: crack, green: spalling. Crack and spalling pixel ratios are annotated above each overlay. Easy samples contain wide, well-separated cracks; Hard samples feature thin, branching cracks with co-occurring spalling.

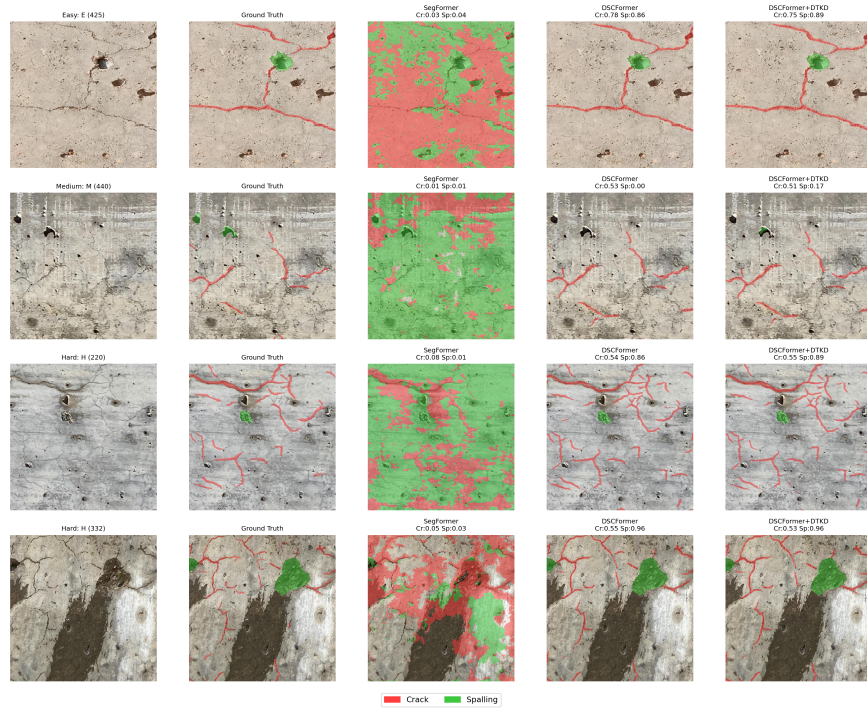


Figure 4: Qualitative comparison on test samples across difficulty tiers. Left to right: input, ground truth, SegFormer-B2, DSCFormer, DSCFormer+DTKD. Per-image crack (Cr) and spalling (Sp) IoU scores are shown above each prediction. Red: crack, green: spalling. SegFormer-B2 exhibits severe over-segmentation; DSCFormer improves localisation; DTKD further refines boundary quality.

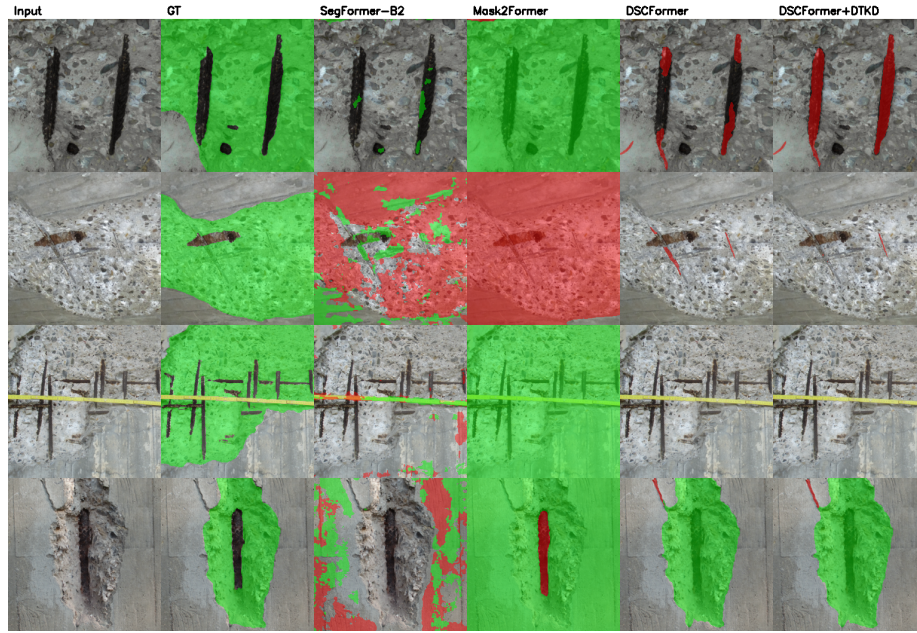


Figure 5: Cross-dataset qualitative results on S2DS (zero-shot). Left to right: input, ground truth, SegFormer-B2, Mask2Former, DSCFormer, DSCFormer+DTKD. Red: crack, green: spalling. SegFormer-B2 and Mask2Former fail on the unseen domain, while DSConv-equipped models retain partial defect localisation.